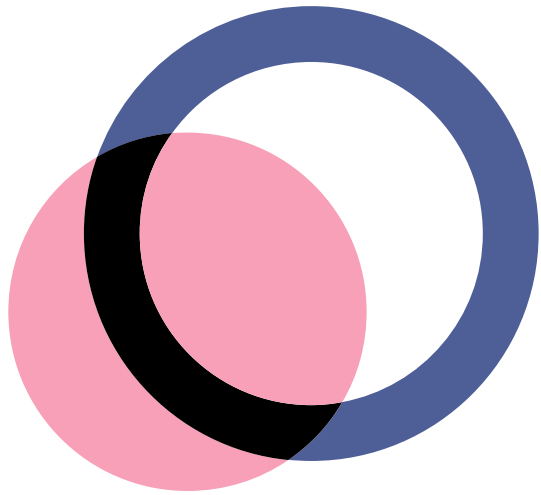
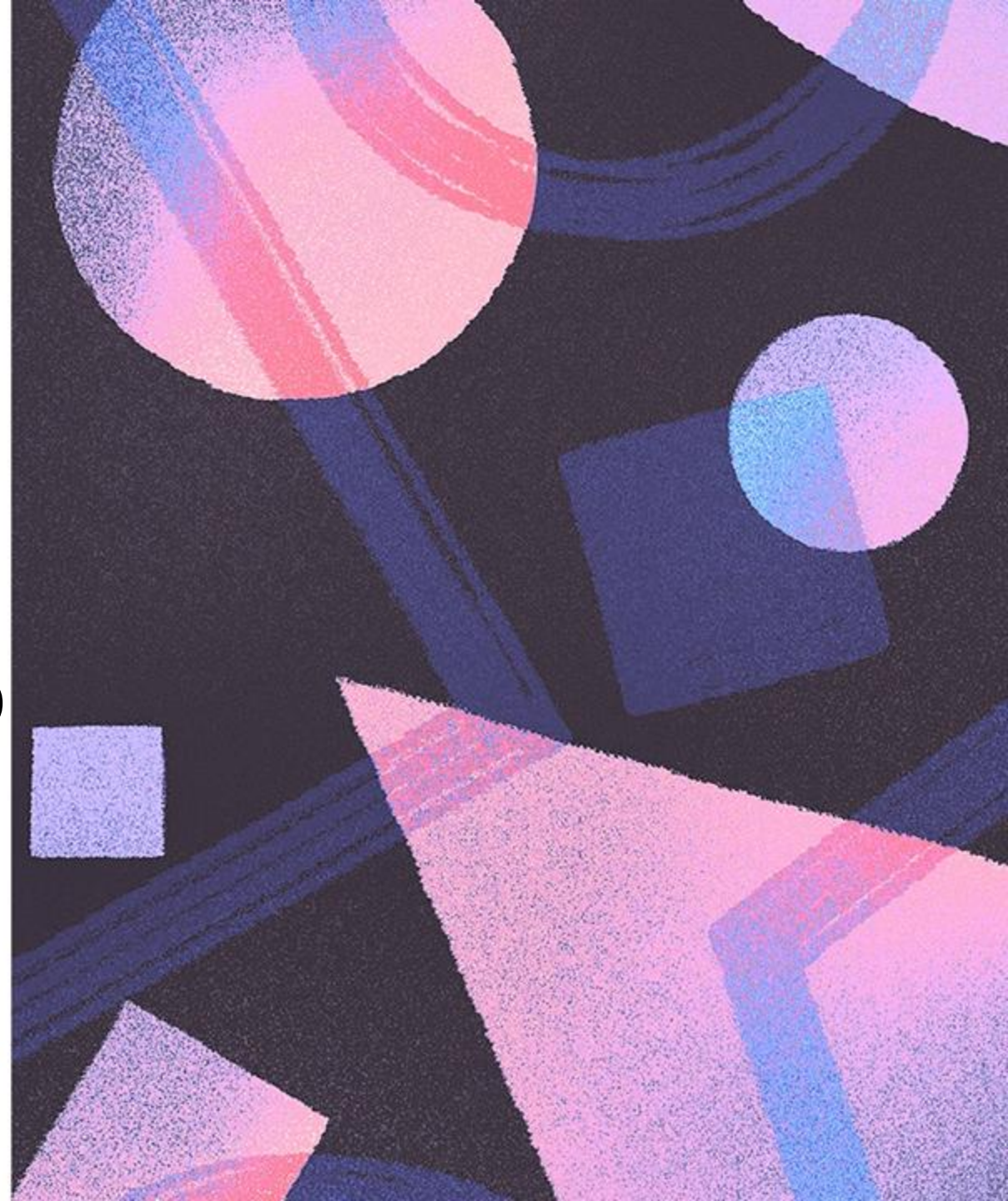




Design Patterns Lab

Lecture 7

> Template Method Pattern



Template Method Pattern

- Part of Behavioral Design Patterns (BDP)
- Focus: Reusable algorithm structure with customizable steps
- Core idea: Fixed process flow + flexible behavior

Problem Statement

- Many systems follow a **fixed sequence of steps**
- But each step may behave differently depending on context
- Challenge:
 - Keep process order consistent
 - Allow customization inside steps
- Need a design that avoids code duplication

Scenario: University Submission System

- System processes student submissions
- Every submission follows the same workflow:
 - Load file
 - Validate format
 - Analyze content
 - Calculate score
 - Store result
- Problem: each department implements steps differently

Template Method Pattern Idea

- Define a **fixed algorithm structure**
- Allow subclasses to define step behavior
- Two key parts:
 - Template method → defines workflow order
 - Primitive methods → customizable steps
- Prevents changing algorithm sequence

Core Design Structure

- Abstract class contains:
 - Template method (final → cannot be changed)
 - Abstract step methods
- Responsibilities split:
 - Parent class → controls workflow
 - Child classes → implement behavior
- Ensures consistency across system

Key Concept: Template Method

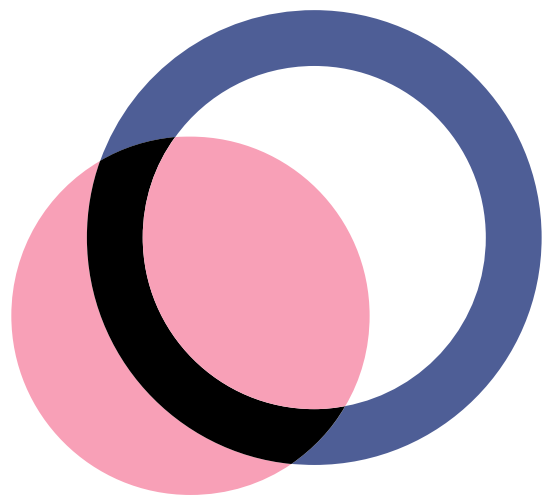
- Central method defines full process flow
- Executes steps in a fixed sequence
- Cannot be overridden by subclasses
- Guarantees algorithm consistency
- Acts as system backbone

Key Benefits

- Code reusability
- Eliminates duplication of workflow logic
- Enforces consistent process execution
- Easy to extend with new departments
- Supports Open/Closed Principle

Activity: Smart Hospital Patient Processing System

- A hospital system processes different types of patient records
- All departments follow the same general workflow
- Each department applies different logic based on patient type (e.g., emergency, regular checkup)
- The goal is to design a system where the process structure stays fixed, but behavior changes per department



**THANK
YOU**

